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Saliency affects attentional capture and suppression of abrupt-onset and color singleton distractors: Evidence from event-related potential studies

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Abstract

Attention is the process of selecting relevant information and suppressing irrelevant information. However, it is still controversial whether attentional capture by salient but task-irrelevant stimuli operates in a bottom-up fashion (stimulus-driven theory) or a top-down fashion (goal-driven theory) or if even salient distractors can be suppressed before capturing attention (signal suppression theory). In the present study, we investigated how saliency affects attentional capture (indexed by N2-posterior-contralateral [N2pc]) and suppression (indexed by distractor positivity [P_D]) of abrupt-onset and color singleton distractors in a visual search task. Experiment 1 showed that an abrupt-onset distractor elicited both N2pc and P_D , while a color singleton distractor elicited only P_D . Moreover, the abrupt-onset distractor elicited a larger N2pc and a larger P_D relative to the color singleton distractor. In addition, both distractors elicited an early positive component, the positivity posterior contralateral (Ppc), which was also larger for abrupt onsets than for color singletons. Experiment 2 further demonstrated that when both the abrupt onset and color singleton were designed as targets, and thus required no attentional suppression, Ppc was elicited, but P_D was not. This corroborated the finding in Experiment 1 that the later P_D , not the early Ppc, reflected attentional suppression. Therefore, a more salient distractor demonstrates stronger early perceptual processing, can capture attention better and needs more attentional resources to be suppressed later. Based on these results, a three-stage hypothesis is proposed, in which the saliency of a distractor modulates processing at early perception, attentional capture, and suppression stages.

KEYWORDS

attentional capture, event-related potentials, N2pc, P_D , Ppc, saliency, suppression

Xiaowen Chen, Bing Xu, and Yanzhang Chen contributed equally to this research.

1 | INTRODUCTION

Attentional capture originally referred to the process of a salient stimulus attracting attention without being restricted by the current task (Yantis & Jonides, 1984). Later studies suggested that object matching of the top-down search goal can also capture attention (Olivers, 2009). There are two opposing views on the mechanism of attentional capture, the bottom-up, stimulus-driven theory (Theeuwes, 1991) and the top-down, goal-driven theory (Folk et al., 1992). The stimulus-driven theory purports that a physically salient stimulus can automatically capture attention and that this type of attentional bias is not subject to top-down cognitive control (Franconeri & Simons, 2003; Hickey et al., 2006; Theeuwes, 1991). In contrast, the goal-driven theory emphasizes the similarity between distractors and the target, and proposes that a distractor having similar features to the search target can capture attention (Folk & Remington, 2010; Remington et al., 1992; Serences et al., 2005). An intermediate theory from Sawaki and Luck (2010), the signal suppression hypothesis, reconciles the contradiction between stimulus-driven and goal-driven theories. Signal suppression theory proposes that a salient stimulus can automatically capture attention, but the bias of attention can be actively inhibited by participants, such that the salient stimulus is suppressed before it can capture attention. That is, if active suppression reacts before attentional capture, the bias to the salient stimulus can be prevented (i.e., consistent with the goal-driven theory); in contrast, if the salient stimulus fails to be suppressed, attentional capture occurs (i.e., consistent with the stimulus-driven theory).

Previous studies on attentional capture have primarily focused on two types of salient stimuli, static (e.g., color, shape, size, orientation, and brightness) and dynamic (e.g., motion, abrupt onset, and abrupt offset) stimuli. Evidence has shown that it is extremely easy to identify dynamic targets among static distractors (Hillstrom & Yantis, 1994; Yantis & Egeth, 1999); static targets could also be found in some kinds of dynamic distractors (e.g., flip or flicker), but when the distractors only changed brightness, the efficiency of searching static targets was reduced (Pinto et al., 2006). Moreover, compared with static distractors, dynamic distractors were demonstrated to increase the interference of ongoing tasks (Franconeri & Simons, 2003; Muhlenen et al., 2005). These findings imply an asymmetry between static and dynamic stimuli in attentional selection.

Color singleton (Kadel et al., 2017; Mertes & Schneider, 2018; Shin & Chong, 2016) and abrupt onset (Folk & Remington, 2015; Gaspelin et al., 2016; Sunny & von Muhlenen, 2013; Yantis & Jonides, 1984) have been the most widely used stimuli in previous research, representing

static and dynamic stimuli, respectively. For color singleton, classic studies have shown that color attribute has a higher influence on search efficiency than shape attribute (Theeuwes, 1991), indicating that color singletons hold relatively higher priority in attentional capture (see also Seiss et al., 2009; Wolfe & Horowitz, 2004). Moreover, task-irrelevant color singletons can also be actively suppressed (Chang & Egeth, 2021; Gaspelin et al., 2017). For abrupt onset, some studies have shown that they were able to capture attention, even when they were presented subliminally (Fuchs et al., 2013), indicating a relatively automatic process (Remington et al., 1992). Evidence suggests that attentional capture and suppression are different between abrupt onset and color singletons. For attentional capture, two lines of evidence suggest that abrupt onset has a higher level of automaticity. First, several studies have shown that task-irrelevant abrupt onset can capture attention but color singleton distractors cannot (Irwin et al., 2000; Jonides & Yantis, 1988; Mounts, 2000). Second, when a color singleton was the target, an abrupt-onset distractor could override the top-down attentional capture of the color singleton (Folk & Remington, 2015; Schreij et al., 2008). For attentional suppression, attentional capture by color singletons can be eliminated or noticeably reduced when participants can adopt a clear attentional control setting (ACS, Folk et al., 1992) that is currently relevant to the target's features. For example, in studies that used additional singleton paradigms, attentional selection was strongly affected by the feature saliency of singletons, which might incidentally lead participants to adopt a broad ACS for detecting any salient features in the visual field (singleton-detection mode). It has been indicated that under this broad ACS, attentional capture by an irrelevant singleton occurs (Bacon & Egeth, 1994). Meanwhile, in search tasks that encouraged a feature-search mode, where distractors were heterogeneous in one feature domain—say, they differed in shape or color—the singleton became less salient, and there is considerable evidence that suppression of the irrelevant color singleton is both possible and robust (Bacon & Egeth, 1994; Chang & Egeth, 2021; Cosman & Vecera, 2013; Gaspelin et al., 2017; Leber & Egeth, 2006). In contrast, evidence is mixed for the idea that attentional capture by abrupt onset is mandatory. There is controversy around whether attentional selection can be immune to the distraction from abrupt onsets. Using a spatial cueing paradigm in which abrupt onset pre-cues could indicate the potential location of either a target or a distractor, both anecdotal and empirical evidence suggests that task-irrelevant abrupt onset is more likely to disrupt the target selection (Jonides & Yantis, 1988; Theeuwes, 1991). However, a substantial body of research has provided conflicting results in which abrupt onset failed to capture attention when it did not

match the current task setting (Schreij et al., 2014) was presented frequently (Folk & Remington, 2015; Neo & Chua, 2006), occurred with practice via a series of training blocks (Turatto & Pascucci, 2016), or was presented under easy tasks instead of difficult tasks (Gaspelin et al., 2016). Nevertheless, few have provided direct evidence that abrupt onsets can be suppressed.

Static and dynamic visual information is processed by parvocellular and magnocellular pathways, respectively (Livingstone & Hubel, 1988; Pinto et al., 2006; Van Essen & Maunsell, 1983). It is well-documented that abrupt onsets are the most powerful attention-grabbing events in the visual sense (Jonides & Yantis, 1988; Yantis & Jonides, 1984), probably because they alert the visual system that a new object has appeared in the scene. This might reflect a general acknowledgment that suppression of abrupt onsets is more difficult than that of color singletons.

The event-related potential (ERP) technique has been used to investigate attentional capture as well as suppression by task-irrelevant stimuli, indexed by N2-posterior-contralateral (N2pc) and distractor positivity (P_D), respectively (Hickey et al., 2009; Luck & Hillyard, 1994a, 1994b). N2pc, usually appearing 180–280 ms after stimuli onset, is the difference wave between ERPs elicited at the contralateral and ipsilateral electrodes, with its maximal amplitude at the PO7/PO8 electrode (Corriveau et al., 2012; Dell'Acqua et al., 2006; Luck & Hillyard, 1994a, 1994b). Studies have shown that N2pc can be elicited by visual features, such as color (Luck et al., 1997), orientation (Conci et al., 2006), size (Luck & Hillyard, 1994a), and movement (Hirai & Hiraki, 2006); by complex visual information, such as geometric figures (Kiss et al., 2008), letters and words (Eimer, 1996); and even social information, such as facial expressions (Holmes et al., 2009) and eye gaze (Doi et al., 2009). Conversely, P_D is considered to be an ERP marker of distractor suppression (Hickey et al., 2009; Sawaki & Luck, 2012), usually with the same scalp distribution as N2pc. However, less consensus has been achieved regarding its time course. There are four main views regarding the time sequence between N2pc and P_D : (1) P_D occurs at roughly the same time window as N2pc (Barras & Kerzel, 2017); (2) P_D occurs before N2pc (Sawaki & Luck, 2010); (3) P_D occurs at the same time window as N2pc and is a subcomponent of N2pc (Hickey et al., 2009); and (4) P_D occurs after N2pc (Kiss et al., 2012). Obviously, the time sequence between N2pc and P_D is still unclear, such that their corresponding processing sequence of attentional capture and suppression also remains to be elucidated.

Abrupt onset and color are two important attributes of stimuli in our environment but may differ in their salience. From an evolutionary standpoint, abrupt onset involves quick changes to the environment, which are

perceptually the same as motion and could be indicative of a greater threat to safety (e.g., from a predator), while static color changes are not as dangerous and are therefore less salient. Several studies have used N2pc and P_D as indicators to separately examine attentional capture or suppression for task-irrelevant abrupt onset or color singleton distractors (Gaspar & McDonald, 2014; Goller et al., 2020; Hickey et al., 2006). For color distractors, previous studies have obtained the following results. First, a color singleton distractor could elicit an N2pc, indicating it captured attention (Hickey et al., 2006; Huang et al., 2016); second, color singleton distractors elicited a P_D immediately after eliciting an N2pc, indicating that it captured attention first, and then was suppressed (Kiss et al., 2012; Sawaki & Luck, 2013, in the long duration condition); third, color singleton distractors directly elicited a P_D without an N2pc in the short duration (200 ms) condition, indicating that it was directly suppressed (Kiss et al., 2012). In contrast, electrophysiological evidence for attentional capture and suppression of abrupt-onset distractors is largely nonexistent. A few studies have shown that abrupt onset failed to elicit an N2pc in some conditions. For example, rare abrupt onset (appearing with a probability of 10% or 20%; Noesen et al., 2014) or abrupt onset that mismatched with target features (Goller et al., 2020) failed to capture attention. Regarding attentional suppression of the abrupt onset, no electrophysiological evidence is yet available. Overall, the existing evidence suggests that the attentional capture and suppression process between color singletons and abrupt onset are different. For attentional capture, task-irrelevant color singletons can elicit an N2pc, whereas the evidence that abrupt-onset distractors can elicit an N2pc is insufficient. For attentional suppression, color singletons can elicit a P_D that represents active suppression, whereas it is still unclear whether abrupt onset can elicit a P_D . So far, no evidence is available for a direct within-subject comparison of attentional capture and suppression between task-irrelevant abrupt onset and color singleton in a single study.

In the present study, we used N2pc and P_D as indicators to directly compare attentional capture and suppression between task-irrelevant abrupt onset and color singleton distractors. We had two main research goals. First, we aimed to explore the difference in attentional capture and suppression between task-irrelevant color singleton and abrupt-onset distractors. Second, we aimed to identify the time course of attentional capture and suppression, specifically, whether P_D occurs before N2pc, after N2pc, or in the same time window as N2pc. In Experiment 1, we hypothesized that color distractors might be suppressed (eliciting a P_D) or they captured attention first and then were suppressed (eliciting both N2pc and P_D), whereas abrupt-onset distractors would capture attention and be suppressed by attention (eliciting

both N2pc and P_D). In Experiment 2, we assumed that P_D would disappear when abrupt onset and color singletons were task-relevant stimuli, as no-suppression process was required for them under this condition. By comparing the P_D between suppression (Experiment 1) and no-suppression (Experiment 2) conditions, we could identify whether suppression (indexed by P_D) occurs before, after, or in the same time window as attentional capture (indexed by N2pc). Importantly, we hypothesized that the difference in amplitudes of N2pc and P_D between abrupt onset and color singleton would indicate the extent of attentional capture and suppression as a function of stimulus saliency. We defined the term “relative saliency” as the difference in stimulus energy between color singletons and abrupt onset. We proposed that stimuli with higher saliency (i.e., abrupt onset) should elicit a larger attentional capture effect and/or attentional suppression effect than stimuli with lower saliency (i.e., color singletons). Work by Ruthruff et al. (2019) has recently supported this assumption. They tried to promote the incentive to suppress abrupt onset via increasing the number of abrupt onsets as well as their stimulus energy (dots were replaced with solid frames). The results show that it was difficult to eliminate or even to significantly reduce the capture effect of abrupt onset. Accordingly, we predicted that a higher-saliency (i.e., abrupt onset) stimulus should elicit larger N2pc and P_D components, relative to a lower-saliency (i.e., color singleton) stimulus.

2 | EXPERIMENT 1: ATTENTIONAL CAPTURE AND SUPPRESSION OF ABRUPT ONSET AND COLOR SINGLETON DISTRACTORS

2.1 | Method

2.1.1 | Participants

Twenty-four students (11 males, $M = 20.08$ years, $SD = 7.08$ years) were recruited from Guangzhou University and participated for a payment of approximately \$10 or course credit. They were naive about the purpose of the experiment. All participants were right-handed, with normal or corrected-to-normal vision and no color blindness reported. Informed consent was obtained from each participant.

2.1.2 | Stimuli and procedure

Stimuli were presented on a 17-inch CRT monitor with a refresh rate of 85 Hz and a resolution of 1024×768 pixels.

Participants were seated in a dimly lit and electrically shielded room, with a viewing distance of approximately 57 cm. The experiment was implemented using E-Prime software (version 3.0, Psychology Software Tools, Inc., Pittsburg, USA). The stimuli in the search array were four equiluminant outlined shapes (23 cd/m^2), two of which were always displayed at a distance of 4.5° above and below the vertical meridian (i.e., the positions at 12 and 6 o'clock), with the additional two stimuli randomly displayed 4.5° relative to the center of the monitor on opposite sides, randomly chosen from six lateral positions (left, upper left, lower left, right, upper right, lower right). Each search stimulus contained a gray dot ($0.2^\circ \times 0.2^\circ$, randomly presented at a distance of 0.35° on the left or right), three of which were displayed in the same shape (either squares or circles); targets were defined as uniquely shaped items that differed from the homogeneous shape distractors (i.e., a square among circles or vice versa). The order of distractor type (color singleton vs. abrupt onset) was randomly assigned in the experiment. In the color singleton conditions, targets and two distractors shared the same colors (red or green, counterbalanced across participants); one of the distractors was defined as a color-unique singleton (red among green or vice versa). Meanwhile, in the abrupt onset conditions, all search items shared the same colors.

The possible location of search items was previewed using several placeholders, which is referred to as the position frame below. Placeholders were combinations of squares and circles ($2^\circ \times 2^\circ$). The number of placeholders depended on the distractors type; specifically, in the color singleton conditions, the possible locations of targets and distractors were preceded by four placeholders, while only three placeholders were given in abrupt onset conditions. The abrupt-onset distractor did not match any locations of these placeholders. Therefore, no prediction of the onset's position could be made ahead of time (see Yantis & Jonides, 1984 for a similar design for stimulus onset). Note that these placeholders did not predict the exact location of search items; they represented search items that were randomly chosen from these preceded locations. These placeholders then altered their shape to predict the type of search items before the search array onset, which is referred to as the alternation frame below. For instance, if the search array consisted of a square target and three circle distractors, placeholders then became one solid square combined with a dotted circle and three solid circles combined with dotted squares or vice versa for the circle targets.

Each trial started with a gray fixation cross ($1^\circ \times 1^\circ$), with its duration randomly selected between 500 and 1000 ms (Figure 1). Participants were instructed to maintain their gaze on the fixation cross, avoiding head and/or eye movements until the end of the trial. This was followed

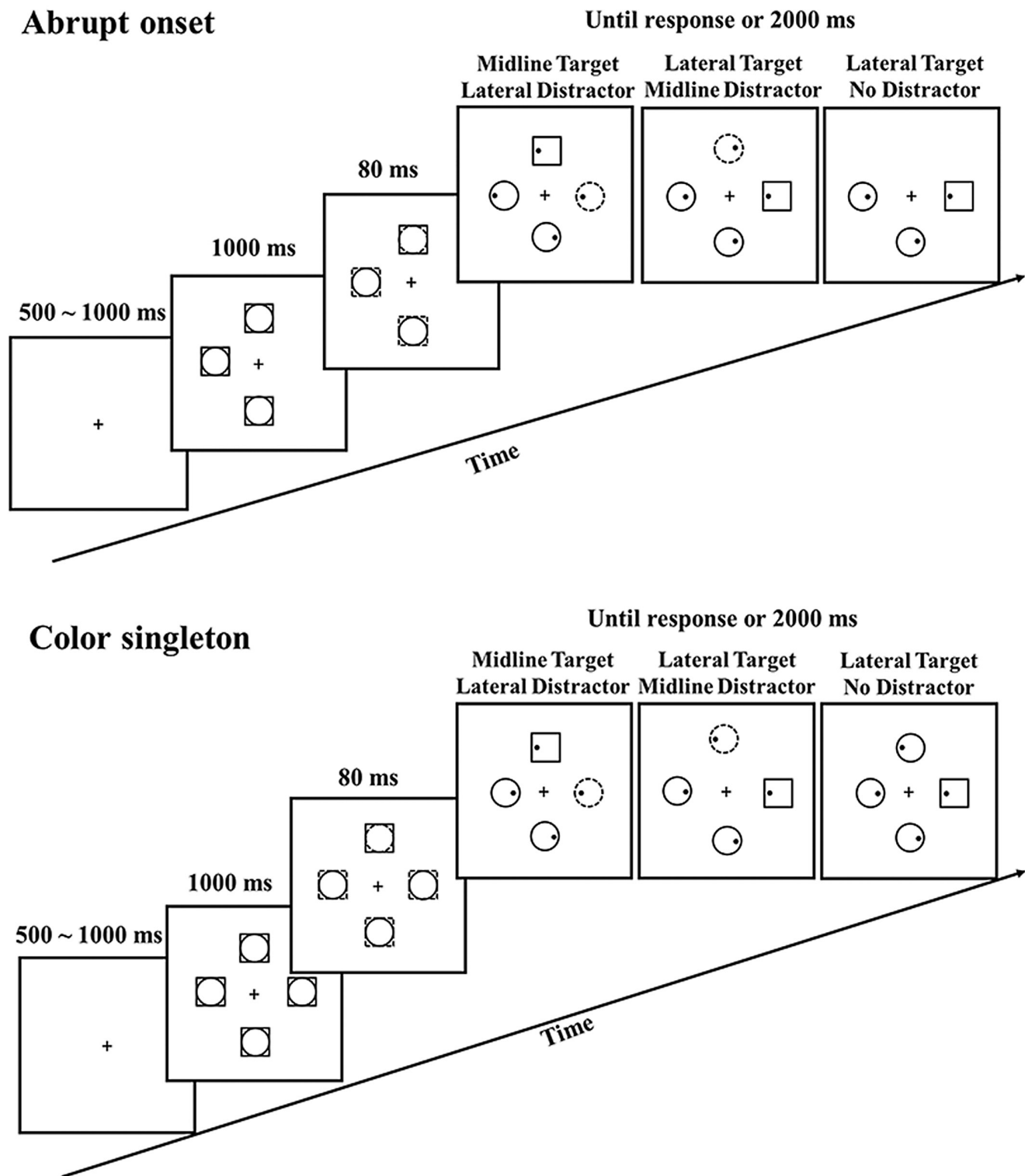


FIGURE 1 Schematic illustration of the procedure in Experiment 1. The upper panel is an illustration of the abrupt onset sequence, and the lower panel is an illustration of the color sequence. In this diagram, the target is a square, while the abrupt onset or color singleton (black dotted circle) is a distractor. Participants were instructed to determine the position of the gray dot in the square (on the left in this illustration). Each sequence has four display types: MTLD, LTMD, LTND, and MTND (see text for details).

by three consecutive frames. The position frame was first presented for 1000 ms, followed by an alternation frame for 80 ms. Finally, the search frame was presented until a

response was detected or 2000 ms had passed. Participants were required to use the “1” or “3” number pad key (counterbalanced across participants) to report the position of

the gray dot inside the target, with equal emphasis on response speed and accuracy.

According to the relative position of targets and distractors, the display type could be divided into four conditions: a lateral target with no distractor (LTND), a midline target with no distractor (MTND), a lateral target with a midline distractor (LTMD), and a midline target with a lateral distractor (MTLD). The experiment was composed of 40 practice trials and 960 formal trials, organized in 24 blocks, with 40 trials for each block. Performance feedback (mean accuracy and reaction times of each block) was presented on the screen during a 30s break after each block.

2.1.3 | Electrophysiological recording and analysis

Electroencephalogram (EEG) data were recorded from an electrode cap with 64 Ag/AgCl electrodes of the extended international 10–20 system, and the EEG signal was recorded with Neuroscan Curry 8 system (Compumedics USA, Charlotte, NC, USA) in AC mode. The continuous EEG signal was sampled at 1000 Hz and filtered with a bandpass of 0.05–400 Hz. Throughout the recording, electrode impedance was maintained below 5 k Ω . Vertical electrooculogram (VEOG) was monitored by placing two electrodes 1.5 cm above and below the left eye. Horizontal electrooculogram (HEOG) was monitored by placing two electrodes 1.5 cm at the external canthi of the left and right eye. The ground electrode was located between FPz and Fz. The online reference electrode was approximately 1.5 cm posterior to Cz, and the offline reference electrode was the average value of the bilateral mastoid.

We used the EEGLAB (Delorme & Makeig, 2004) and ERPLAB (Lopez-Calderon & Luck, 2014) MATLAB toolbox to analyze EEG data. ERPs elicited by the target were measured in LTND and LTMD conditions, while ERPs elicited by the distractor were measured only in the MTLD condition. The MTND condition was excluded in later statistical analysis because there was neither target nor distractor in the lateral position. EEG data were bandpass filtered offline with 0.1 Hz high-pass and 30 Hz low-pass. EEG data were segmented into epochs starting from 200 ms before to 600 ms after the search array onset. Epochs were baseline corrected by subtracting the average signal recorded in the 200 ms before stimulus onset. After excluding trials associated with an incorrect response in the visual search task, individual trials were then rejected if a step function detected thresholds of 30 μ V for HEOG, 50 μ V for VEOG, and 80 μ V for all channels. This procedure led to four subjects being excluded due to more than 30% of trials being rejected. For the remaining 20

subjects, the average percentage of rejected trials was 10.7% (range = .3–27.7%).

For the ERPs time-locked to the search screen, we computed the contralateral and the ipsilateral portions of the N2pc and P_D (i.e., the average between PO7 activity elicited by a right presented target/distractor and PO8 activity elicited by a left presented target/distractor for the former, and vice versa for the latter). Mean amplitudes of N2pc and P_D were computed by subtracting the ipsilateral waveforms from the contralateral waveforms. To estimate the time window of these components, a collapsed localizers approach was used (Gaspelin, 2017), in which the ERP waveform was averaged across abrupt-onset trials and color singleton trials and the measurement window was defined as the onset and offset of these components in this averaged difference wave (the latencies at which the voltage reached 50% of the peak amplitude). The time window used was 340–440/240–270 ms for N2pc for targets/distractors and 300–400 ms for P_D for distractors.

The mean latency of the subtracted N2pc and P_D components was estimated using the jackknife-based approach combined with a relative criterion technique (Kiesel et al., 2008) to calculate the time point when each jackknife subsample reached 50% fractional peak latency. The *t* and *p* values were corrected according to the formula [$t_{\text{adjusted}} = t/(n - 1)$] (Ulrich & Miller, 2001).

We performed analyses of behavioral and EEG data in R (R Core Development Team, 2017) by using the ezANOVA function in the “ez” package (Lawrence, 2011). A Greenhouse–Geisser correction was used if the assumption of sphericity was violated. Estimation of ERLs were also compared by means of mixed models. Each analysis returned a Bayes Factor (*BF*) using the “BayesFactor” package (Rouder et al., 2012), which implements the Jeffreys–Zellner–Siow (JZS) default prior on effect sizes (Rouder et al., 2012). The *BF* ratio quantified the likelihood of data fitting under the *H*₁ (alternative hypothesis) versus the *H*₀ (null hypothesis), which reflected the strength of evidence within the data that was in favor of our hypothesis. Unfortunately, due to the absence in the literature of a proposal to correct the *BF* estimated with jackknifed data, it was not possible to confirm this result with mixed models.

2.2 | Results

2.2.1 | Behavioral results

Only correct responses were included in analyses, and trials with RTs less than 200 ms or greater than 1500 ms and RTs exceeding three standard deviations above/below the mean for each participant and condition were

considered outliers and excluded (5.55% of the total data). The analysis of RTs only included distractor present and distractor-absent conditions. That is, the LTMD and MTLD conditions were collapsed together as the distractor present condition, and the MTND and LTND conditions were collapsed together as the distractor-absent condition. Table 1 shows the accuracy (ACC) and reaction times (RTs) of Experiment 1 in the four testing conditions (Figure 1).

A one-way repeated-measures ANOVA with distractor types (abrupt onset, color singleton, on distractor) as within-subject factors was conducted. The results (as shown in Figure 2) showed that the main effect of distractor types was significant [$F(2, 38) = 31.19, p < .001, \eta_p^2 = 0.621, BF_{10} > 1000$] and RTs were slower in abrupt-onset trials (590 ms, $p < .001$, Cohen's $d = 2.24$) and color singleton trials (588 ms, $p < .001$, Cohen's $d = 2.08$) compared to distractor-absent trials (564 ms).

Further, the results of ACC showed that the main effect of the distractor types was not significant [$F(2, 38) = 0.92, p = .409, \eta_p^2 = 0.046, BF_{01} = 3.82$].

2.2.2 | ERP results

Target analysis

N2-posterior-contralateral. Figure 3 illustrates the grand-average contralateral and ipsilateral ERP waveforms

TABLE 1 ACC and RTs (standard error) in Experiment 1.

	ACC (%)		RTs (ms)	
	Abrupt onset	Color	Abrupt onset	Color
Absent	96.32 (0.46)	95.84 (0.53)	553 (10)	574 (10)
Present	95.50 (0.62)	95.50 (0.58)	585 (10)	586 (10)

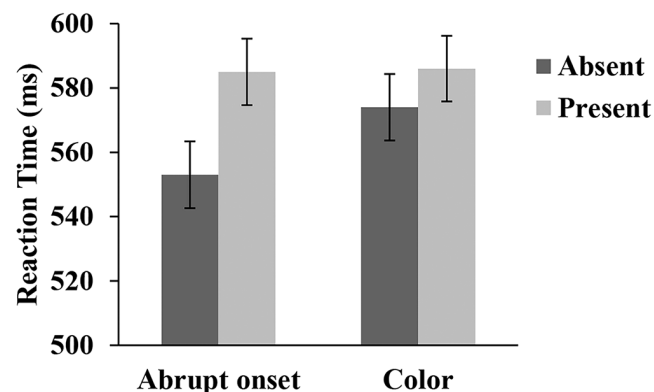


FIGURE 2 Mean RTs for trials with or without abrupt onset or color singleton distractors. Error bars show the standard error of the mean RTs.

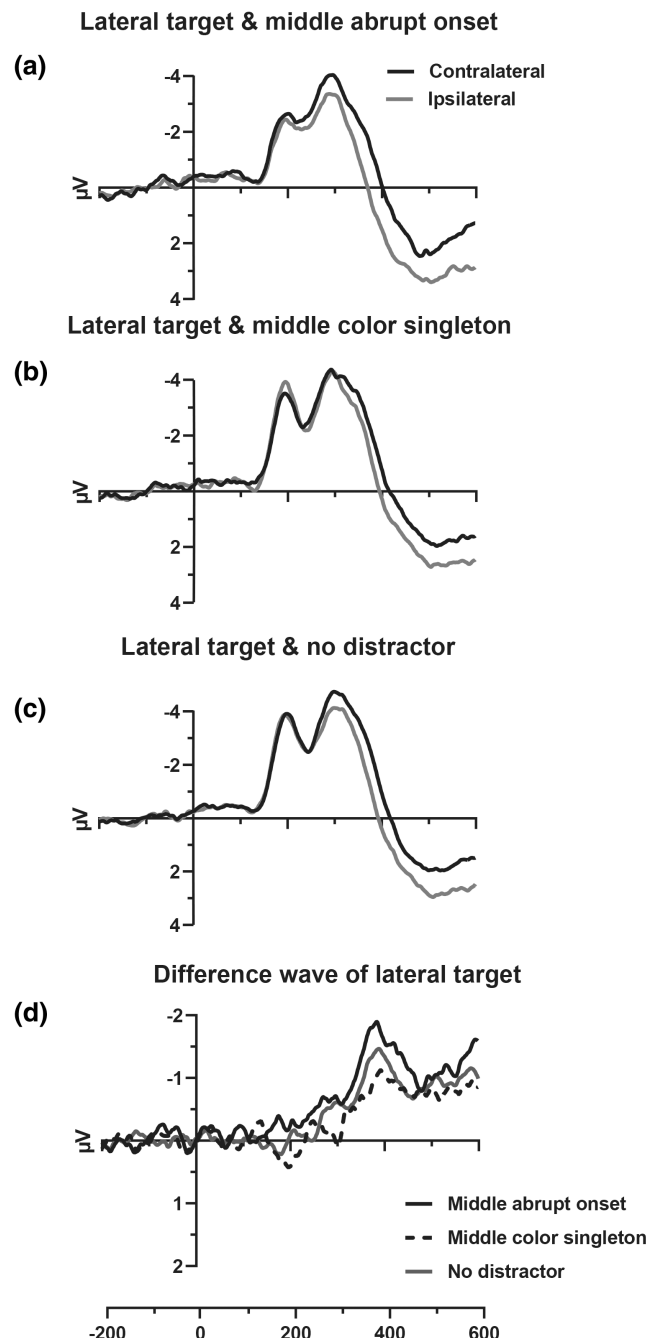


FIGURE 3 Electrophysiological results from Experiment 1 for targets. The upper diagrams show grand-averaged contralateral and ipsilateral ERPs elicited by lateral targets for abrupt-onset trials (a), color singleton trials (b), and no distractor trials (c) at PO7/PO8. The lower diagrams (d) show difference waves, which were calculated by subtracting the ipsilateral ERPs from the contralateral ERPs, for abrupt onset, color singleton, and no distractor, respectively.

recorded at PO7/8 elicited by the lateral target in different distractor conditions. To determine the effect of midline distractors on N2pc when the target was lateralized, ERP mean amplitudes elicited by lateral targets were submitted to a 3×2 repeated-measures ANOVA, including

distractor types (abrupt onset vs. color singleton vs. no distractor) and laterality (contralateral vs. ipsilateral) as within-subject factors. Results showed greater negativity at contralateral sites compared to ipsilateral sites in all distractor types [$F(1, 19) = 7.22, p = .015, \eta_p^2 = 0.275, BF_{10} = 580.17$], suggesting that a reliable N2pc was present in all conditions. The main effect of distractor type was significant [$F(2, 38) = 8.86, p = .005, \eta_p^2 = 0.318, BF_{10} = 36.97$], but the interaction between the two factors was not significant [$F(2, 38) = 1.49, p = .238, \eta_p^2 = 0.073, BF_{01} = 5.14$]. The lack of interaction further emphasized that the target's N2pc was not affected by type of distractors (abrupt onset: $-1.56 \mu V$, Cohen's $d = -0.64$; color singleton: $-0.88 \mu V$, Cohen's $d = -0.36$; no distractor: $-1.15 \mu V$, Cohen's $d = -0.47$).

2.3 | Distractor analysis

2.3.1 | N2-posterior-contralateral

Figure 4 shows the grand-average ERP waveforms elicited by lateral distractors at electrodes PO7/PO8 when targets were displayed along the vertical midline. To determine the lateralized effect of distractors, mean amplitudes during 240–270 ms were submitted to a 2×2 repeated-measures ANOVA, including distractor types (abrupt onset vs. color singleton) and laterality (contralateral vs. ipsilateral) as within-subject factors. The results showed that the main effect of distractor types was not significant [$F(1, 19) = 0.05, p = .833, \eta_p^2 = 0.002, BF_{01} = 4.21$], while the main effect of laterality was significant [$F(1, 19) = 10.15, p = .005, \eta_p^2 = 0.348, BF_{10} = 2.43$]. Importantly, the interaction between distractor types and laterality was significant [$F(1, 19) = 16.35, p < .001, \eta_p^2 = 0.462, BF_{10} = 9.98$], suggesting that abrupt onset elicited a larger N2pc, relative to color singleton. Pairwise comparisons showed the presence of a reliable N2pc only for the abrupt-onset distractor ($-1.37 \mu V, p < 0.001$, Cohen's $d = -1.56, BF_{10} = 63.60$) but not for the color singleton distractor ($0.12 \mu V, p = .546$, Cohen's $d = 0.13, BF_{01} = 3.63$). Scalp distribution maps for the N2pc elicited by abrupt onset and color singleton distractors are shown in Figure 5 (middle panels).

2.3.2 | Distractor positivity

Mean amplitudes obtained during 300–400 ms were submitted to a 2 (distractor type: abrupt onset vs. color singleton) $\times 2$ (laterality: contralateral vs. ipsilateral) repeated-measures ANOVA. This analysis revealed a main effect of laterality, indicating that the contralateral ERP was more positive, relative to the ipsilateral ERP,

demonstrating a reliable P_D was present [$F(1, 19) = 72.38, p < .001, \eta_p^2 = 0.792, BF_{10} > 1000$]. More importantly, there was an interaction of distractor types $\times$ laterality [$F(1, 19) = 52.97, p < .001, \eta_p^2 = 0.736, BF_{10} = 55.67$]. Pairwise comparisons between contralateral and ipsilateral ERP amplitude showed a larger P_D for abrupt onset, relative to the color singleton distractor (abrupt onset: $2.71 \mu V, p < .001$, Cohen's $d = 2.89, BF_{10} > 1000$; color singleton: $0.86 \mu V, p < .001$, Cohen's $d = 0.92, BF_{10} = 150.09$). Scalp distribution maps for the P_D elicited by abrupt-onset and color singleton distractors are shown in Figure 5 (right panels). An additional test was performed to explore whether the onset latency of P_D differed between distractor types. Planned comparison revealed that P_D was elicited earlier for abrupt onset than for color singleton distractors [305 vs. 349 ms, $t_{\text{adjusted}}(19) = -4.10, p_{\text{adjusted}} < .001$].

2.3.3 | Early contralateral positivity

Figure 4 also shows that an early contralateral positivity was elicited before the N2pc. The time window for estimating this component determined by the collapsed localizers approach was 160–215 ms. A 2×2 repeated-measures ANOVA with the same factors as the P_D analysis revealed that the main effects were significant for both distractor types [$F(1, 19) = 16.58, p < .001, \eta_p^2 = 0.466, BF_{10} > 1000$] and laterality [$F(1, 19) = 78.82, p < .001, \eta_p^2 = 0.806, BF_{10} > 1000$]. Additionally, the interaction between target type and laterality was also significant [$F(1, 19) = 87.79, p < .001, \eta_p^2 = 0.822, BF_{10} = 33.85$]. This suggests that the amplitude of early contralateral positivity was larger for abrupt onsets than for color singletons (1.91 vs. $0.54 \mu V$). Pairwise comparisons confirmed that reliable contralateral positivity was elicited by abrupt onsets ($p < .001$, Cohen's $d = 3.09, BF_{10} > 1000$) as well as by color singletons ($p = .001$, Cohen's $d = 0.88, BF_{10} = 31.07$). Scalp distribution maps for this early contralateral positivity elicited by abrupt onsets and color singleton distractors are shown in Figure 5 (left panels). Planned comparison revealed that the onset latency of this early contralateral positivity did not differ between abrupt onsets and color singletons [161 vs. 189 ms, $t_{\text{adjusted}}(19) = 0.34, p_{\text{adjusted}} = .74$].

2.4 | Discussion of Experiment 1

Experiment 1 was conducted to explore the difference in attentional capture and suppression between abrupt-onset and color singleton distractors. The results were clear; the behavioral data showed a robust capture effect when both types of distractors were presented. However, attentional responses were different between abrupt onset and color

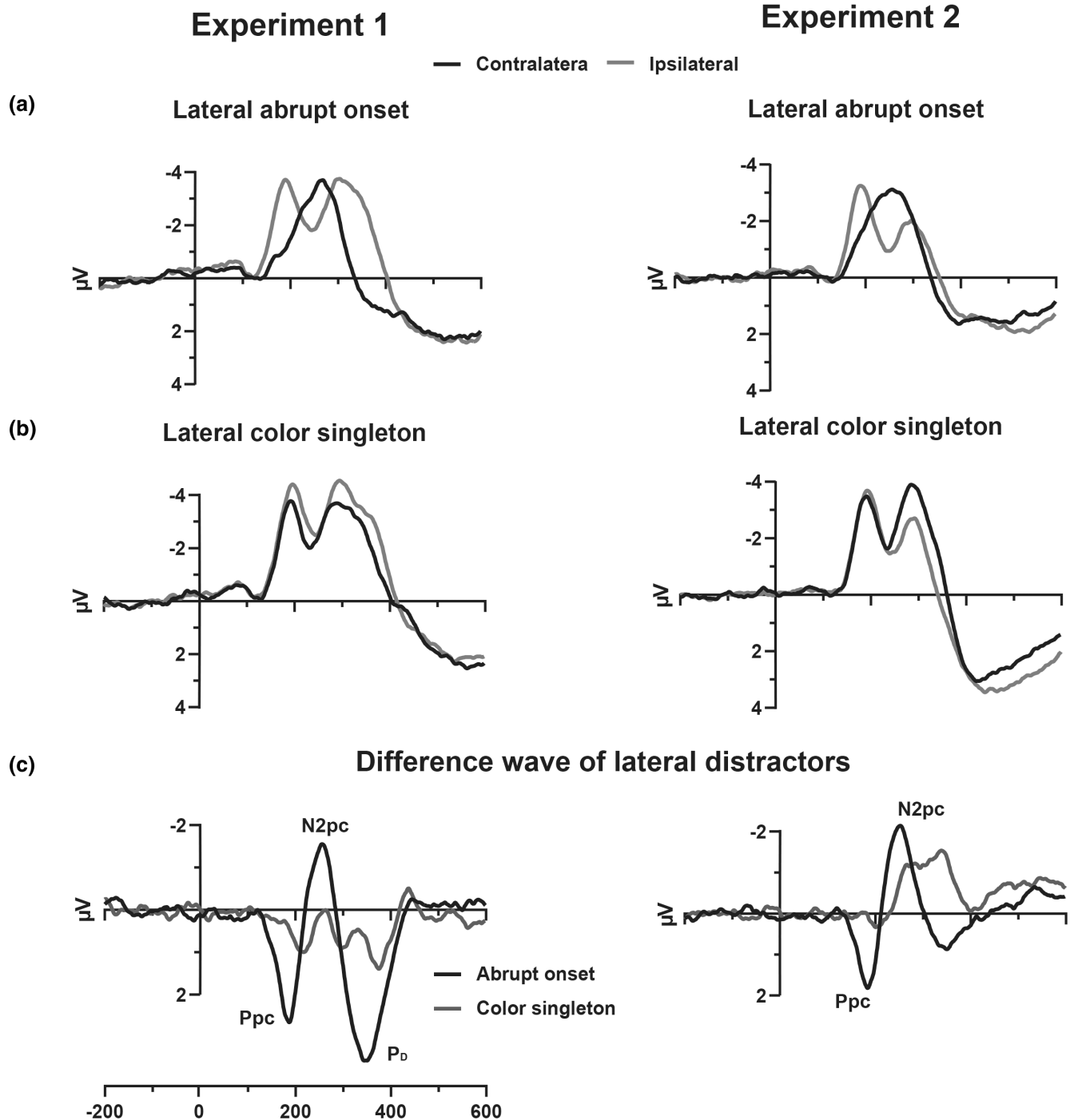


FIGURE 4 Electrophysiological results from Experiment 1 for distractors (abrupt onset and color singleton) and from Experiment 2 for target (abrupt onset and color singleton). The upper diagrams show grand-averaged contralateral and ipsilateral ERPs for abrupt onset (a) and color singleton (b) at PO7/PO8. The lower diagrams (c) show difference waves, which were calculated by subtracting the ipsilateral ERPs from the contralateral ERPs, for abrupt onset and color singleton, respectively. As can be seen in the figure, when both abrupt onset and color singleton were used as distractors (Experiment 1), N2pc was elicited only for the abrupt onset; when both were targets (Experiment 2), neither of them elicited a P_D.

singleton. We found that N2pc occurred before P_D when search targets were accompanied by abrupt onset. This finding revealed that attentional selection was biased toward the abrupt onset first and then disengaged from it. Given that we expected that, with abrupt onsets, it would

be difficult to eliminate the effect of attentional capture, we implicitly argued that the suppression of abrupt onsets was active. Perhaps more surprisingly, while behavioral data showed significant RT cost when color singletons were presented compared to when they were absent, ERPs

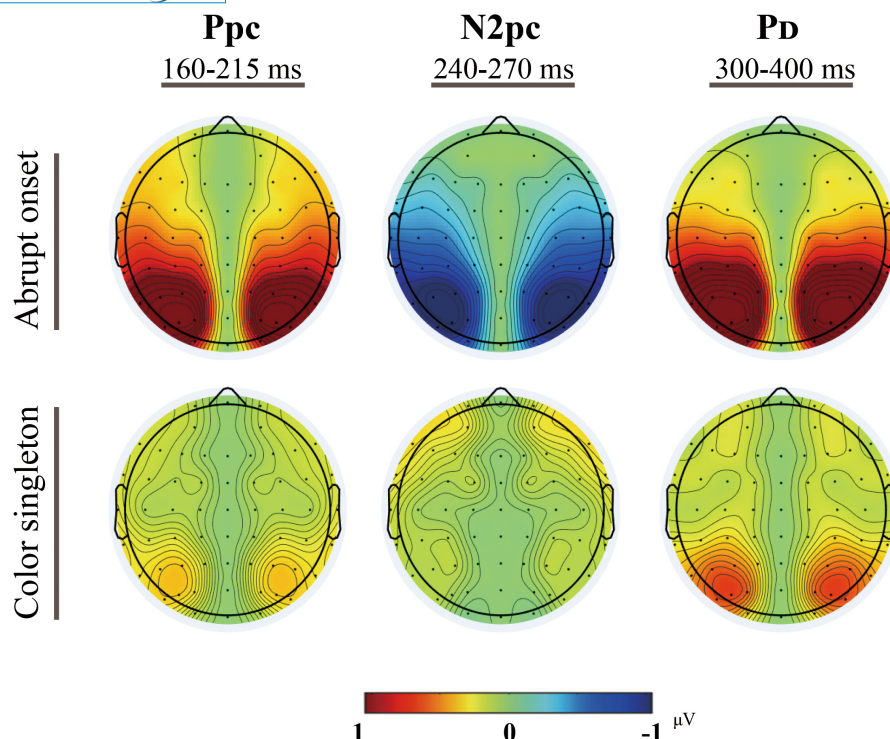


FIGURE 5 Grand mean topographical distribution of the difference wave by subtracting the ipsilateral ERPs from the contralateral ERPs for abrupt onset (upper panel) and color singleton (lower panel) distractors in Experiment 1. Three components, Ppc (left-most panels), N2pc (middle panels), and P_D (right-most panels) are plotted, mirrored in both hemispheres. As can be clearly seen in the figure, abrupt-onset distractor elicited a larger Ppc, N2pc, and P_D, relative to color singleton distractor.

elicited by the lateral color singletons did not trigger the N2pc but P_D only. In the present task, the searched distractors were homogeneous, which clearly encouraged a broad ACS for detecting any salient features, namely singleton-detection mode (Bacon & Egeth, 1994); it is known that under this ACS, attentional capture by an irrelevant singleton occurs.¹ The following discussion focuses on this challenging finding.

Considering that participants can fully anticipate the type of distractors by the preceding number of placeholders (i.e., three placeholders predicted the abrupt-onset distractors, four placeholders predicted the color singleton distractors), it is tempting to assume that the absence of N2pc in color singleton trials was due to proactive suppression (see also Gaspar & McDonald, 2014; Sawaki & Luck, 2010, for color singletons eliciting only P_D and not N2pc). In this case, one may expect color singletons to elicit a suppression-related P_D component earlier than abrupt onsets, even being observed in the typical N2pc time window. However, the jackknife-based approach combined with a relative criterion technique to compare the P_D onset latency between abrupt onsets and color singletons did not provide any support

for this possibility. We found that P_D emerged earlier and larger in abrupt-onset trials than in color singleton trials, suggesting that attentional suppression to abrupt onset was faster and stronger than that to color singletons. Alternatively, distractor interference indexes the competition exerted by the color singleton when present but not necessarily that attention was shifted to its location. The spatial filters of the target and color singleton work in parallel. As the attentional response to color singletons (P_D) overlapped with the attentional response to targets (N2pc), two polarities reversed components, thereby neutralizing each other, thereby attenuating or even eliminating the N2pc. This type of neutralization should have occurred primarily in color singleton trials, where rejecting color-unique distractors and searching for shape-unique targets were different attentional task sets, but not in the abrupt-onset trials, where targets and all distractors were uniformly colors. When collectively taking the findings of the target's ERPs, if the deployment of attention toward color singletons and abrupt onsets is automatic, the N2pc for lateralized targets should presumably be delayed and/or reduced in its magnitude when additional singletons are presented. Instead, we found that N2pc amplitudes for lateralized targets were not affected by the presence of distractors. Taken together, although our

¹We would like to thank an anonymous reviewer for these helpful comments.

behavioral results indicated that the color singleton did capture attention, the absence of N2pc in color singleton trials did not provide direct evidence of proactive suppression.

The results shown in Figure 4 indicate that both types of distractors elicited an early contralateral positivity, with a larger magnitude for abrupt onsets than for color singletons. This positivity posterior contralateral (Ppc) component is a putative index of initial, bottom-up response to the presence of a salient stimulus (Barras & Kerzel, 2017; Fortier-Gauthier et al., 2012). This finding suggests that early perception is stronger for abrupt onsets than for color singletons. We thus proposed the role of “relative saliency” that may affect the attentional response to task-irrelevant distractors. This may explain, at least in part, the absence of N2pc in color singleton trials. Presumably, the magnitude of Ppc may represent the degree of perceptual sensitivity in an initial early stage of information processing that would be followed by a more refined attentional selection leading to the N2pc for covert attention. In this case, when the more pronounced abrupt onset is present in early perception, and in contrast to situations where color singletons are static, a larger capture effect will be observed; therefore, suppression enhancement may have to be employed to bias the subsequent attentional selection away from abrupt onsets. Consequently, the more salient the distractor, the higher the extent of attentional capture and suppression.

We also note that the nature of the Ppc is still under debate. There are two alternative explanations of this early contralateral positivity. First, Kerzel and Burra (2020) proposed that this early contralateral positivity was the initial N2pc that indexed the attentional deployment to the lateral item, followed by the later N2pc to a salient color singleton on the opposite side. Their rationale was that participants had a lateral (horizontal) advantage for scanning, relative to vertical stimuli, as they responded faster to targets on the horizontal display. However, results from the present study do not support this hypothesis. We found that participants took the same time to identify the lateral target and vertical target when there was no distractor [564 vs. 567 ms, $t(19) = -0.42$, $p = .679$, Cohen's $d = 0.121$, $BF_{01} = 4.30$] or when the distractor was present (583 vs. 588 ms, $p = .494$, Cohen's $d = 0.141$, $BF_{01} = 4.21$).

Second, this component was named P_D , reflecting earlier attentional suppression in another study (Gaspelin & Luck, 2018a). Recently, following the challenging finding of Kerzel and Burra (2020), Drisdelle and Eimer (2021) proposed that the early contralateral positivity was the first P_D , reflecting attentional suppression to the lateral salient color singleton, and the component following the

contralateral negativity was a second P_D elicited by the opposite non-salient distractor.

Therefore, while we tentatively named this early contralateral positivity as Ppc and assumed it reflected the early perception of salient distractors in Experiment 1, Experiment 2 was conducted to identify whether attentional suppression occurs before or after attentional capture (i.e., to identify the time sequence of N2pc and P_D). Specifically, the abrupt onset and the color singleton were configured as search targets in Experiment 2. In this case, participants should have no incentive to suppress their attentional deployment. If the Ppc in Experiment 1 was a real P_D , then P_D occurred earlier than N2pc, indicating an earlier suppression; otherwise, if P_D occurred after N2pc, then attentional suppression occurs after attentional capture. Moreover, if the Ppc in Experiment 1 was a real P_D , then it should disappear in a task requiring no attentional suppression; otherwise, it should be unaffected by task relevance, or at least, exist in a task requiring no attentional suppression. In other words, if the early contralateral positivity was a Ppc reflecting relative saliency among search items, as we proposed, we should replicate this finding of Experiment 1 regardless of task relevance.

3 | EXPERIMENT 2: ATTENTIONAL CAPTURE AND SUPPRESSION OF THE ABRUPT ONSET TARGET AND THE COLOR SINGLETON TARGET

3.1 | Method

3.1.1 | Participants

Twenty students (9 males, $M = 21.64$ years, $SD = 3.03$ years) from the same pool as in Experiment 1 participated for payment or course credit. Informed consent was obtained from each participant.

3.1.2 | Stimuli and procedure

The stimuli and procedure of Experiment 2 were the same as those used in Experiment 1, except that both the abrupt onset and color singleton distractors became task-relevant stimuli. Specifically, if three placeholders were presented in the positioning screen, the target was the abrupt onset stimulus; if four placeholders were presented in the positioning screen, the target was the color singleton stimulus. Participants performed 36 practice trials, followed by 270 trials for each condition, for a total of 576 trials.

3.1.3 | Electrophysiological recording and analysis

The same criteria used in Experiment 1 were applied in Experiment 2. The artifact rejection procedure led to three subjects being excluded due to more than 30% of trials being rejected. For the remaining 17 subjects, the average percentage of rejected trials was 6.9% (range = .4%–17.9%).

The time window selected for Ppc was 160–200 ms. A jackknife-based approach combined with the relative criterion technique on the N2pc onset latency revealed that N2pc emerged significantly earlier for abrupt onset targets than for color singleton targets [224 vs. 251 ms, $t_{\text{adjusted}}(16) = 3.24$, $p_{\text{adjusted}} = .005$]. Therefore, the time window selected for N2pc was 220–320 ms for abrupt onset targets and 250–350 ms for the color singleton targets.

3.2 | Results

3.2.1 | Behavioral results

Incorrect responses, trials with RTs less than 200 ms or greater than 1500 ms, and RTs exceeding three SDs were rejected from further analysis (6.78% of the total data). Table 2 shows the accuracy (ACC) and reaction times (RTs) of Experiment 2.

A paired-sample t test confirmed that there was no significant difference in ACC between abrupt onset and color singletons [$t(16) = 0.98$, $p = .339$, $BF_{01} = 2.03$]. No RT difference was found as well [$t(16) = 1.56$, $p = .138$, $BF_{01} = 2.53$], indicating that abrupt onset target and color target had no significant difference in task difficulty.

3.2.2 | ERP results

Positivity posterior contralateral

As shown in Figure 3c, contralateral positivity emerged earlier for the abrupt onset target than for the color singleton target. To determine the effect of early contralateral positivity, ERP mean amplitudes during 160–200 ms were submitted to a 2×2 repeated-measures ANOVA, including the within-subject factors of target type (abrupt onset, color singleton) and laterality (contralateral, ipsilateral). The results showed that main effects were significant for both target types [$F(1, 16) = 43.57$, $p < .001$, $\eta_p^2 = 0.731$,

$BF_{10} > 1000$] and laterality [$F(1, 16) = 30.81$, $p < .001$, $\eta_p^2 = 0.658$, $BF_{10} = 820.53$]. More importantly, the interaction between target types and laterality was also significant [$F(1, 16) = 50.80$, $p < .001$, $\eta_p^2 = 0.760$, $BF_{10} = 134.89$], suggesting stronger early positivity for the abrupt onset target, relative to the color singleton target. Pairwise comparisons between contralateral and ipsilateral ERP amplitude confirmed that a reliable Ppc was elicited by the abrupt onset target ($1.47 \mu\text{V}$, $p < .001$, Cohen's $d = 2.54$, $BF_{10} > 1000$) but not the color singleton target ($0.09 \mu\text{V}$, $p = .413$, Cohen's $d = 0.16$, $BF_{01} = 2.95$).

N2-posterior-contralateral

An analogous 2×2 ANOVA was carried out for the N2pc elicited by abrupt onset targets and color singleton targets. The results showed only a main effect of laterality [$F(1, 16) = 24.27$, $p < .001$, $\eta_p^2 = 0.603$, $BF_{10} > 1000$], suggesting that both abrupt onset and color singletons elicited an N2pc. The BF_{01} was 2.06 for the effect of target type and 2.74 for the interaction of target type and laterality. These results provide critical support for the statistical equivalence of N2pc amplitudes elicited by abrupt onset and color singletons (-1.05 vs. $-1.24 \mu\text{V}$).

3.3 | Discussion of Experiment 2

Experiment 2 investigated the time sequence of attentional capture and suppression by identifying whether the early contralateral positive component was a Ppc or a P_D. The critical finding was that the early contralateral positivity still existed when the abrupt onset was the target, suggesting that the early component in Experiment 1 was a Ppc instead of a P_D. Planned comparison across Experiments 1 and 2 for the Ppc elicited by abrupt onsets revealed that their difference did not reach statistical significance [1.91 vs. $-1.47 \mu\text{V}$, $t(31.70) = 1.64$, $p = .111$]. This further emphasized that this early contralateral positivity was independent of task relevance. This finding is in line with work by Jannati et al. (2013), who found that a salient stimulus could elicit a Ppc, regardless of its task relevancy. Thus, the Ppc observed in the present study was more likely to reflect perceptual processing related to salient stimuli (e.g., Fortier-Gauthier et al., 2012), instead of early attentional deployment (Kerzel & Burra, 2020) or early attentional suppression (Drisdelle & Eimer, 2021).

Using the same manipulation as in our Experiment 2, whereby color singletons became the target, Gaspelin and Luck (2018b) found that the early contralateral positivity disappeared, and therefore, they ruled out the possibility that the early contralateral positivity was due to salience. Although we do not have an explanation for the minimal Ppc activity when color singletons became

TABLE 2 ACC and RTs (standard error) in Experiment 2.

Stimulus type	ACC (%)	RTs (ms)
Abrupt onset	95.15 (0.53)	499 (7)
Color singleton	94.08 (0.79)	509 (10)

the search target in Experiment 2, visual inspection of Figure 3 and the corresponding analysis indicated that abrupt onset targets elicited N2pc earlier than color singleton targets; this is in line with our expectation that higher saliency stimuli are processed stronger and earlier. Similarly, Barras and Kerzel (2017) found that color singletons elicited a Ppc when participants were searching for a shape-defined target, whereas shape singletons did not elicit a Ppc when targets were defined by color. This asymmetry in eliciting Ppc is likely due to color stimulus being more salient than shape stimulus. That is, when the saliency of distractor is high relative to target, distractor can elicit the Ppc. In line with this inference, in the present study, the higher-saliency abrupt onset elicited a larger Ppc relative to the lower-saliency color singleton. Presumably, the amplitude of Ppc may reflect not only the saliency of target stimuli in the visual field but also the relative saliency between the distractor and the target. Nevertheless, recently work by Kerzel & Cong (2021) have found that with closer spatial proximity of distractors to the target location, comes smaller Ppc amplitude to the lateral singleton distractor, suggesting that it does reflect attentional processing to some degree. On the other hand, the Ppc was equally present for their singleton target and singleton distractor, suggesting that the Ppc reflects some task independent processes, as in the present study. Conclusions from Kerzel & Huynh Cong (2021) suggest that Ppc is simultaneously affected by task demands and saliency, and it is difficult to decide “*whether the Ppc reflects bottom-up or top-down effects.*” Future work may be needed to elucidate the nature of Ppc component, and to investigate this interesting analogy between abrupt onsets and color singletons.

Two lines of evidence from the present study support that attentional suppression over abrupt onset occurred after attentional capture. First, the early Ppc observed reflected early perceptual processing that was not a P_D , because it was present in both experiments, regardless of task relevance (i.e., requiring suppression or not). Second, the late P_D was a real P_D , because it was absent in Experiment 2, which required no suppression. That is, the present results suggest that P_D occurred after N2pc and Ppc occurred before N2pc. Thus, our results are consistent with the findings of Kiss et al. (2012) that attentional suppression occurs after attentional capture. However, our results do not support the notion of early lateral attention bias (Kerzel & Burra, 2020) or early suppression (Drisdelle & Eimer, 2021).

Perceptual processing, as indicated by the Ppc component, was stronger for abrupt onset compared to color singletons. As proposed above, this is likely due to their difference in saliency. According to the hypothesis of the

saliency map (Koch & Ullman, 1985), the most salient locations guide subsequent attention; Ppc might be interpreted as a sign of saliency on the map (Krummenacher et al., 2014). Consistent with this hypothesis, previous studies examining Ppc found that color singletons elicited a larger Ppc, relative to shape singletons (Barras & Kerzel, 2016), suggesting that the saliency was stronger for color, relative to shape. Taken together, the present findings suggest a hierarchical structure of stimulus saliency, with saliency declining from abrupt onset, color singleton, and shape singleton in turn.

In short, the Ppc, N2pc, and P_D components in Experiments 1 and 2 suggest a three-stage hypothesis of attentional processing for abrupt onsets. The first is perceptual processing as indicated by Ppc, the second is attentional capture, as indicated by N2pc, and the third is attentional suppression, as indicated by P_D . The attentional difference between the abrupt onset and the color singleton in these three stages might be accounted for by their relative saliency.

4 | GENERAL DISCUSSION

In the present study, we investigated differences of attentional capture and suppression between abrupt-onset and color singleton stimuli, as well as the time sequence of attentional capture and suppression. We had three main findings: (1) Attentional suppression over abrupt-onset distractors is reactive, as evidence shows that abrupt-onset distractors elicit N2pc first, followed by P_D ; (2) abrupt-onset stimuli elicit Ppc, regardless of task relevance; and (3) saliency affects the processing of abrupt onset and color singleton at early perceptual (as indexed by Ppc), attentional capture (as indexed by N2pc), and late suppression (as indexed by P_D) stages. Overall, these results support a three-stage hypothesis of attentional response to abrupt onset: the first stage is perceptual processing, followed by attentional capture, and finally, attentional suppression.

4.1 | Stronger attentional capture and suppression for the abrupt onset, relative to the color singleton distractor

The present results show differences between abrupt onset and color singleton distractors in attentional capture and suppression. Specifically, the abrupt-onset distractor captured attention first and then was suppressed (i.e., with both N2pc and P_D), while color singleton distractors elicited P_D only, providing no direct evidence in support of proactive suppression (i.e., with only P_D). Previous studies

suggested that the occurrence of attentional capture or suppression might depend on the difficulty of search tasks (Feldmann-Wustefeld & Schubo, 2013; Kerzel et al., 2018; Ruthruff et al., 2020). In particular, the suppression of a distractor occurs in simple tasks, while the attentional capture of distractor occurs in difficult tasks. In simple tasks, targets “pop out” more and are easily detected (Wolfe, 1994), such that a distractor is more likely to be suppressed directly. In contrast, in difficult tasks, distractors have high similarity with targets, so they are more likely to capture attention. The task in the present study was easy because the target was a uniquely shaped object. The relatively short RTs and high ACCs in Experiment 1 confirmed this point. Thus, it was expected that both the abrupt onset and color singleton distractors should be directly suppressed in the simple task. However, the present results showed an N2pc for abrupt-onset distractors, indicating that abrupt-onset distractors biased attentional selection regardless of the top-down ACS. Thus, we believe that whether a distractor captures attention or is suppressed depends not on the task difficulty but on the relative saliency between the target and the distractor. In other words, if the saliency of the distractor is higher than that of the target, the distractor might capture attention first before being suppressed later.

In addition, the present results showed that abrupt onset could be suppressed after attentional capture, which was different from the results presented by Ruthruff et al. (2019). They found that abrupt onset was not suppressed using indirect behavioral indicators, because the attentional capture of abrupt onset did not disappear or decrease. The reason for these differing results might be that we used P_D as an indicator of suppression, whereas Ruthruff et al. (2019) measured the attentional capture effect.

4.2 | Saliency affects the attentional capture and suppression of the distractor

The difference between abrupt onsets and color singletons in attentional capture and suppression might be attributed to their difference in saliency. Consistent with this, several behavioral studies have shown that abrupt-onset distractors can capture attention based purely on saliency, while color singleton stimuli capture attention only when they are task-relevant (Folk & Remington, 2015; Liao & Yeh, 2013; Ruthruff et al., 2020). Moreover, the more salient the stimulus, the earlier and stronger the attentional capture. For instance, an ERP study showed that a singleton color target elicited N2pc earlier than a categorical target and non-singleton color, indicating that the singleton color target was more salient (Callahan-Flintoft

& Wyble, 2017). The present study showed that the abrupt-onset distractor captured attention earlier and was stronger relative to the color singleton distractor, indicating that the abrupt onset was more salient than the color singleton. It seems that saliency can affect the process of attentional capture and suppression of the abrupt onset and color singleton distractors.

Moreover, the present study indicated that the more salient the distractor, the stronger the attentional capture and suppression. Consistent with our results, several behavioral studies have shown that there is a positive correlation between the saliency of a stimulus and the extent of its attentional capture and suppression (Donk & van Zoest, 2008; Failing & Theeuwes, 2020; Theeuwes, 1992). In the attentional capture stage, abrupt-onset distractors capture more attention (as indexed by an N2pc only for abrupt onset), relative to color singleton distractors. In the attentional suppression stage, the abrupt-onset distractor has a stronger and earlier suppression process (as indexed by a larger and earlier P_D), relative to the color singleton distractor.

4.3 | Attentional capture occurs before attentional suppression

Within the current literature, the time sequence of attentional capture and suppression remains controversial. There are three main views regarding this issue: (1) suppression occurs at the same time as attentional capture (Barras & Kerzel, 2017; Hickey et al., 2009); (2) suppression occurs before attentional capture (Sawaki & Luck, 2010); and (3) suppression occurs after attentional capture (Kiss et al., 2012). The present study showed that P_D for abrupt onsets was elicited after N2pc, indicating that suppression occurred after attentional capture. Similarly, Kiss et al. (2012) found the same result that distractors could elicit an N2pc followed by a P_D . In other words, abrupt onsets must capture attention before they can be suppressed (Beck et al., 2018).

However, signal suppression theory holds that salient stimuli can be suppressed before attentional capture, showing that color singletons do not elicit N2pc (225–300 ms), but instead elicit an earlier P_D (115–225 ms; Sawaki & Luck, 2010). This result indicates that suppression might occur before attentional capture, which is different from the present result. This difference might be due to the existence of two different types of suppression, namely reactive suppression, and active suppression (Gaspelin & Luck, 2019). Our results support reactive suppression, which means that salient distractors first capture attention, and then are suppressed by attention. Alternatively, whether the early P_D observed in the study by Sawaki and

Luck (2010) is a real P_D , or is similar to the Ppc component in the present study deserves further investigation. In fact, we observed an early contralateral positivity, the Ppc component for abrupt onset in both experiments (similar to the time range of the P_D in Sawaki & Luck, 2010); importantly, however, Experiment 2 identified that the Ppc was not a P_D that reflects suppression, because it appeared even when the task required no suppression.

4.4 | A three-stage hypothesis of attentional responses to abrupt onsets

Based on the present results, we propose a three-stage hypothesis of the attentional responses to abrupt onsets, including early perceptual, attentional capture, and suppression stages, as indexed by Ppc, N2pc, and P_D components, respectively (Figure 6). In this model, each stage is independent, indispensable, and its processing sequence cannot be modified. First, each stage is functionally independent of one another and has its own processing window. Second, each stage is indispensable, which means that abrupt-onset distractors should go through three stages of early perception, attentional capture, and suppression. Although color singletons did not elicit a significant N2pc, they still went through the process of attentional capture; in fact, behavioral results showed longer RTs and lower ACCs when color singletons were used as distractors, indicating the existence of attentional capture. Third, the processing sequence of the three stages cannot be changed, with early perceptual processing occurring before attentional capture and suppression, and attentional capture occurring before suppression. This hypothesis is inconsistent with the signal suppression theory, which proposes that suppression occurs before attentional

capture; however, it supports the “search and destroy” hypothesis (Moher & Egeth, 2012), which proposes that distractors capture attention before being suppressed.

Moreover, in this three-stage hypothesis, the saliency of distractors influences processing at all three stages. First, the higher the saliency of a distractor, the stronger the process of early perception, attentional capture, and suppression. Second, our hypothesis holds a similar viewpoint to the recent version of signal suppression theory (Luck et al., 2021). That is, attentional responses toward a salient distractor depend on whether the gain of this salient distractor can be proactively decreased (Luck et al., 2021). The signal suppression theory holds a positive opinion on the possibility of active suppression (Sawaki & Luck, 2010), but it is probably not always possible to suppress highly salient objects, such as abrupt onsets in the present study. The current three-stage hypothesis stipulates that the bottom-up saliency of a distractor is also a key factor that affects attentional processing. As proposed by Luck et al. (2021), the key factor in determining successful active suppression is “whether proactive control of feature gain can be strong enough to overcome high levels of salience, not whether proactive control of feature gain is possible at all.” In other words, whether a distractor can capture attention or be suppressed depends on the relative strength between the top-down or goal-directed suppression and the bottom-up or stimulus-driven saliency. If the bottom-up saliency is stronger, the distractor will capture attention; if the top-down suppression is stronger, the distractor will be suppressed directly. Third, it should be noted that the three-stage hypothesis may only apply to salient distractors, such as abrupt onset; however, if the distractor is less salient than a color singleton, such as a shape singleton, it may be directly suppressed without attentional capture. That is, the three-stage hypothesis is not

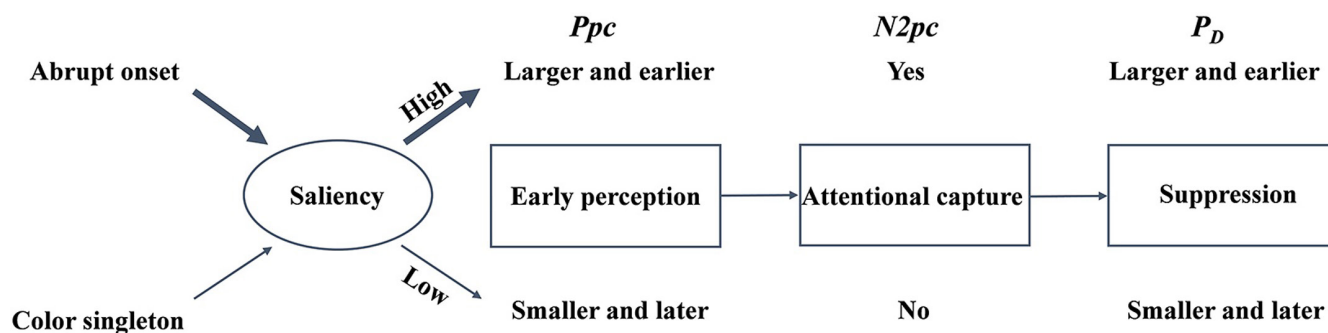


FIGURE 6 Three-stage hypothesis of attentional processing based on abrupt onset and color singleton distractors. The rectangular boxes represent the processing stages, while the circle represents the modulatory factor. In this hypothesis, abrupt onset and color singleton distractors go through all three stages of early perception (as indexed by Ppc), attentional capture (as indexed by N2pc), and suppression (as indexed by P_D). The three stages are independent, indispensable, and the processing sequence cannot be changed (see text for details). Moreover, the saliency of stimuli has an effect on each stage. Specifically, an abrupt-onset distractor has a higher saliency, such that it elicits larger and earlier Ppc, a significant N2pc, and a larger and earlier P_D . In contrast, a color singleton distractor has a lower saliency, such that it elicits a smaller and later Ppc, no significant N2pc, and a smaller and later P_D .

necessarily mutually exclusive with signal suppression theory (Luck et al., 2021; Sawaki & Luck, 2010).

4.5 | Limitations

The current study has several limitations that should be addressed. First, the three-stage hypothesis may be limited to specific high-saliency stimuli, such as abrupt onset and color singleton distractors. It is unclear whether the three stages are still indispensable and unchangeable when the distractor is less salient, such as a shape singleton distractor. Besides, because we did not test the assumption of three-stage in a search task that discourages capture (i.e., feature-search), we assume that this pattern of results can only explain data under singleton-detection mode so far. Second, the present study cannot exclude the possibility of active suppression that depends on sufficient experience with the distractor (Gaspelin & Luck, 2019). Future studies can examine whether the N2pc elicited by a salient distractor decreases or disappears after extensive exercise or experience. Third, the behavioral results were not consistent with the ERP results in Experiment 1. Specifically, behavioral results in Experiment 1 of the present study showed that both abrupt onset and color singleton could capture attention, whereas ERP results showed that only abrupt onset elicited N2pc. More evidence is needed to explain the inconsistency in behavior and ERP results. Finally, Experiment 2 was aimed at identifying the role of Ppc by configuring both color singletons and abrupt onset as search targets in the same situation as that of Experiment 1. We assumed that the presence of the Ppc component in Experiment 2 should reflect the relative saliency instead of attentional suppression. This is critical for determining whether the attentional suppression over salient distractors is proactive or reactive. Nevertheless, as the behavioral measures failed to reflect the relative saliency between abrupt onsets and color singletons, one may expect a similar N2pc amplitude between these two types of targets. This leads to a potential limitation in Experiment 2, where the general procedure was the same as in Experiment 1; the shape target therefore became a shape singleton and may have competed with the target and even smeared the contralateral portion of the N2pc elicited by lateral targets. As far as this situation is concerned, one previous study from Kiss et al. (2008) showed that contingent capture operates at the level of stimulus dimensions. That is, when the dimension of distractors did not match the current target dimension, say, a color target accompanied by a shape singleton, N2pc was small but nearly significant when the target's location was unpredictable (Experiment 1) and entirely absent when the target's location was predictable (Experiment 2). It must

be stressed, however, that it could be the case that this happened in both color singleton target trials and abrupt onset target trials, so we argued that the difference between Ppc and N2pc should account mainly for the difference in their relative saliency (stimulus energy). Future work will need to clarify the role of relative saliency in target selection via a more comprehensive design.

In summary, the abrupt-onset distractor had stronger attentional capture (as indexed by N2pc) and suppression (as indexed by P_D), relative to the color singleton distractor, presumably due to its higher saliency. Attentional capture occurs before suppression, and early perceptual processing (as indexed by Ppc) occurs before attentional capture. A three-stage hypothesis, including early perception, attentional capture, and suppression stages, is proposed for the attentional processing of abrupt onsets, in which all processing stages are mandatory and cannot be bypassed, with a fixed processing sequence in which attentional capture always occur before suppression and early perceptual processing always occurs before attentional capture and suppression. Moreover, the saliency of a distractor affects the processing at all stages, with a higher-saliency distractor eliciting stronger processing at the stages of early perception (i.e., a larger and earlier Ppc), attentional capture (i.e., a larger and earlier N2pc), and suppression (i.e., a larger and earlier P_D), relative to lower-saliency distractors. In short, the attentional processing of a salient distractor is a result of a competition between bottom-up, salience-induced attentional capture and top-down, goal-directed suppression, with a more salient distractor eliciting stronger early perception, capturing more attention, and being harder to suppress later.

AUTHOR CONTRIBUTIONS

Xiaowen Chen: Conceptualization; formal analysis; investigation; visualization; writing – original draft. **Bing Xu:** Data curation; formal analysis; project administration; validation; writing – original draft. **Yanzhang Chen:** Formal analysis; investigation; validation; visualization; writing – original draft. **Xianqing Zeng:** Data curation; software; visualization. **Yue Zhang:** Data curation; validation; writing – original draft. **Shimin Fu:** Conceptualization; funding acquisition; investigation; methodology; project administration; resources; supervision; writing – review and editing.

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DATA AVAILABILITY STATEMENT

Data available on request from the authors.

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